

Module 05: Action in High School

Learning Objectives

1. Define **Action** within the context of High School.
2. Analyze how Action interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Action.

Introduction

This module explores **Action**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Action is a critical component of the 8-part Active Inference spine, bridging the gap between Cognition and Learning.

Key Concepts

1. Action as a Markov Blanket Boundary

How does Action define the boundary between the agent and the environment?

2. Generative Models of Action

What parameters involved in Action must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Action drive the perception-action loop?

Applications

In High School, we see Action manifest in: * **Specific Example 1:** A robot vacuum like Roomba performs Active Inference action: it predicts clear floor space using its bump sensors

and infrared cliff detectors, and when it hits an unexpected obstacle (prediction error), it immediately changes direction -- its action policy is to follow the path that minimizes the expected surprise of collisions, gradually building a map of the room through accumulated prediction errors. * **Specific Example 2:** Autocorrect on your phone is an action system driven by Active Inference: the keyboard predicts what word you intend to type based on a language model and your typing patterns, and when the prediction error between your actual keystrokes and the predicted word exceeds a threshold, it takes the action of substituting the most probable correction -- which is why it sometimes "corrects" unusual words you actually intended to type.

Conclusion

Understanding Action allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.